

Fear in the mouse brain

If you have experienced a jump scare, and then continued to be frightened for a while, it is a type of persistent internal state. It might just be the shadow of a curtain moving, or an unexpected noise, and even after you identify the source, your senses remain on high alert, your may start sweating and your heart beats faster. The persistent reaction is similar to how a mouse feels when it is introduced to a threat which is then taken away, say a rat. Researchers from Caltech have investigated what goes on in a mouse brain on a molecular level, when the brain is in a persistent state of fear. The researchers found that the electrical signals spiked in a primitive portion of the brain, called the



hypothalamus which is associated with reflexive actions, as against deliberate thoughts or memory,

which are carried out in the cortex. The research shows that the hypothalamus is not always only reflexive in action, and that it is capable of more long term processing. The researchers then investigated how the neurons in the hypothalamus could stay active for much longer than they could physically be fired for. Proposed scenarios include a sequential relay, or a chemical introduced into the environment that causes the neurons to keep firing, or both. <http://dgit.in/msfear>

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This month in Earth we explore climate history, in Space Age we learn more about intermediate mass black holes, in Health we learn about new scientific developments in the fight against COVID-19, in Ignorance we find out why there's no such thing as a fish, and much more.

The Hype Machine

A new book by MIT professor Sinan Aral, called The Hype Machine explores why social media suddenly took over, how it is changing the world, and the steps that can be taken to fix it. The human brain has evolved as a response to social signals to such an extent that we feel better when we are connected, which is why people seem to be addicted to social media. Studies showed that a large number of likes triggers more dopamine,

a neurotransmitter associated with pleasure. Another study showed that fake news items get stronger reactions on social



media, because they are both novel and polarising, at least more so than what real news can be. For this reason, fake news spreads faster on social

media. Platforms have to follow a business model that encourages the sharing of dubious content, as these attract more engagement. These platforms work in what is called the attention economy. The steps that can be taken to improve the situation include automated and user generated tags for posts, as well as mechanisms to reduce revenue collection from dubious pieces of content. <http://dgit.in/hypem>



Torn between stars

All the planets in the Solar System orbit along the same plane. Astronomers have now found a three star system where the accretion disc around the stars has been warped and twisted. Scientists now hope to find planets in the system, which are likely to have highly oblique orbits. <http://dgit.in/torndisc>



Your PC in space

GomSpace is launching a satellite to demonstrate propulsion and communication technologies, but piggybacking on GomX-4B is a circuit board with cheap off the shelf electronics, to check how well conventional computer parts hold up to the harsh environment of outer space. <http://dgit.in/flashspace>



Stellar wind captured

Astronomers have captured direct images of the stellar winds that surround a star just before they go nova, providing unprecedented insight into their deaths. The bubbles of the stellar winds are not spherical, as they were so far believed to be, but shaped by nearby stars and planets. <http://dgit.in/stellarwind>



Dragonfly delayed

The Dragonfly mission to the Saturnian moon of Titan to find potential signs of life, has been delayed by a year from 2026 to 2027. The delay was due to a number of factors, including the COVID-19 outbreak. The Dragonfly mission is a multi rotor drone that can cover more area than a rover. <http://dgit.in/latedragon>